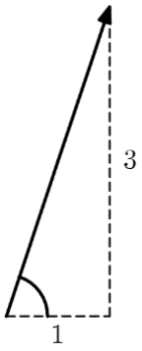
1.68 Forces: Vectors

Worked Solutions

1 a)

$2\mathbf{i} - \mathbf{j} - \mathbf{i} + 4\mathbf{j} = \mathbf{i} + 3\mathbf{j}$

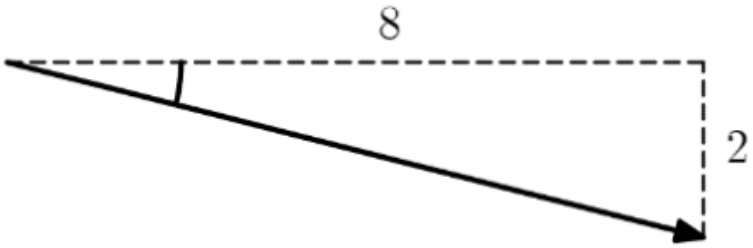


Resultant force: $\sqrt{1^2 + 3^2} = \sqrt{10} = 3.16$ N (3 SF)

Angle with Ox: $\theta = \tan^{-1}\left(\frac{3}{1}\right) = 71.6°$ (3 SF)

1 b)

$\binom{3}{2} + \binom{5}{-4} = \binom{8}{-2}$



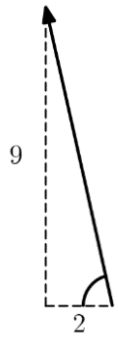
Resultant force: $\sqrt{8^2 + 2^2} = 2\sqrt{17} = 8.25$ N (3 SF)

Angle with Ox: $\theta = -\tan^{-1}\left(\frac{2}{8}\right) = -14.0°$ (3 SF)

346° (3 SF)

1 c)

$3\mathbf{i} + \mathbf{j} - 5\mathbf{i} + 8\mathbf{j} = -2\mathbf{i} + 9\mathbf{j}$

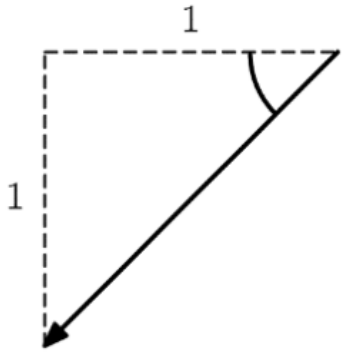


Resultant force: $\sqrt{2^2 + 9^2} = \sqrt{85} = 9.22$ N (3 SF)

Angle with Ox: $\theta = 180° - \tan^{-1}\left(\frac{9}{2}\right) = 103°$ (3 SF)

1 d)

$\binom{-5}{-9} + \binom{4}{8} = \binom{-1}{-1}$



Resultant force: $\sqrt{1^2 + 1^2} = \sqrt{2} = 1.41$ N (3 SF)

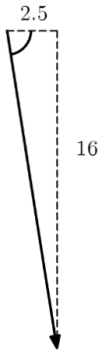
Angle with Ox: $\theta = -\left[180° - \tan^{-1}\left(\frac{1}{1}\right)\right]$

$= -135°$ (3 SF)

$225°$ (3 SF)

1 e)

$-0.5\mathbf{i} + 2\mathbf{j} + 3\mathbf{i} - 18\mathbf{j} = 2.5\mathbf{i} - 16\mathbf{j}$



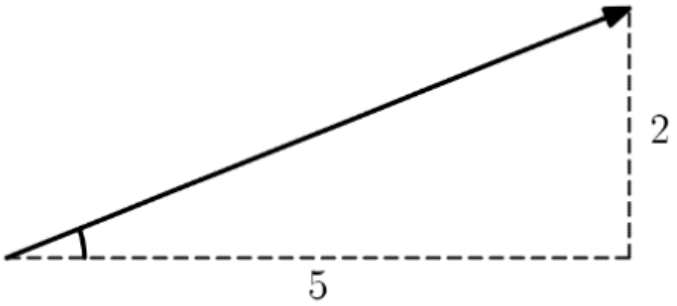
Resultant force: $\sqrt{2.5^2 + 16^2} = 16.2$ N (3 SF)

Angle with Ox: $\theta = -\tan^{-1}\left(\frac{16}{2.5}\right) = -81.1°$ (3 SF)

279° (3 SF)

2 a)

$5\mathbf{i} + 3\mathbf{j} - \mathbf{j} = 5\mathbf{i} + 2\mathbf{j}$



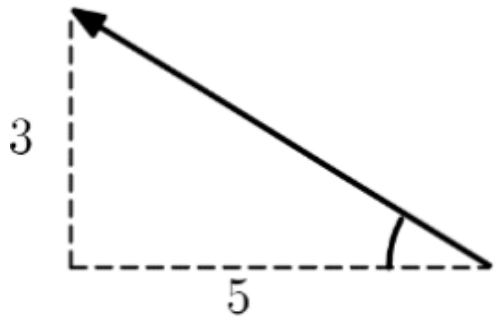
Resultant force: $\sqrt{5^2 + 2^2} = \sqrt{29} = 5.39$ N (3 SF)

$\theta = \tan^{-1}\left(\frac{2}{5}\right) = 21.8014\ldots°$

Bearing: 068°

2 b)

$\binom{-5}{2} + \binom{0}{1} = \binom{-5}{3}$



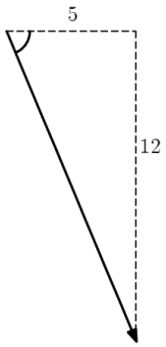
Resultant force: $\sqrt{5^2 + 3^2} = \sqrt{34} = 5.83$ N (3 SF)

$\theta = \tan^{-1}\left(\frac{3}{5}\right) = 30.96375653\ldots^\circ$

Bearing: 301°

2 c)

$2\mathbf{i} - \mathbf{j} + 3\mathbf{i} - 11\mathbf{j} = 5\mathbf{i} - 12\mathbf{j}$



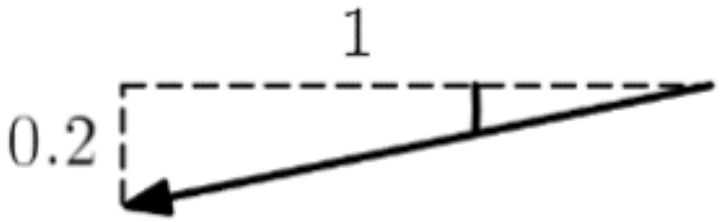
Resultant force: $\sqrt{5^2 + 12^2} = 13$ N

$\theta = \tan^{-1}\left(\frac{12}{5}\right) = 67.380\ldots^\circ$

Bearing: 157°

2 d)

$\binom{5}{-1.2} + \binom{-6}{1} = \binom{-1}{-0.2}$



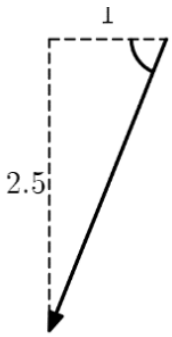
Resultant force: $\sqrt{1^2 + 0.2^2} = 1.02$ (3 SF)

$\theta = \tan^{-1}\left(\frac{0.2}{1}\right) = 11.30993\ldots^\circ$

Bearing: 259°

2 e)

$1.25\mathbf{i} - 0.75\mathbf{j} - 2.25\mathbf{i} - 1.75\mathbf{j} = -\mathbf{i} - 2.5\mathbf{j}$



Resultant force: $\sqrt{1^2 + 2.5^2} = 2.69$ N (3 SF)

Angle with Ox: $\theta = \tan^{-1}\left(\frac{2.5}{1}\right) = 68.19859\ldots^\circ$

Bearing: 248°

3)

$\sqrt{k^2 + 25} = \sqrt{29} \Rightarrow k = 2$

4)

$\sqrt{4k^2 + 4k^2} = \sqrt{8k^2} = 2\sqrt{2}k = 12$

$\Rightarrow k = \pm 3\sqrt{2}$ So $k = -3\sqrt{2}$

5 a)

$\mathbf{P} + \mathbf{Q} = -2\mathbf{i} + \mathbf{j} - 3\mathbf{i} + 11\mathbf{j} = -5\mathbf{i} + 12\mathbf{j}$

Force required for equilibrium: $5\mathbf{i} - 12\mathbf{j}$

5 b)

$\mathbf{P} + \mathbf{Q} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \begin{pmatrix} -4 \\ \sqrt{3} \end{pmatrix} = \begin{pmatrix} -5 \\ \sqrt{3} \end{pmatrix}$

Force required for equilibrium: $\begin{pmatrix} 5 \\ -\sqrt{3} \end{pmatrix}$

5 c)

$\mathbf{P} + \mathbf{Q} = \sqrt{2}\mathbf{i} - \mathbf{j} - 3\mathbf{j} = \sqrt{2}\mathbf{i} - 4\mathbf{j}$

Force required for equilibrium: $-\sqrt{2}\mathbf{i} + 4\mathbf{j}$

5 d)

$\mathbf{P} + \mathbf{Q} = \begin{pmatrix} k \\ 1 \end{pmatrix} + \begin{pmatrix} -k \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$

Force required for equilibrium: $\begin{pmatrix} 0 \\ -2 \end{pmatrix}$

5 e)

$\mathbf{P} + \mathbf{Q} = a\mathbf{i} + b\mathbf{j} - a\mathbf{i} + 2b\mathbf{j} = 3b\mathbf{j}$

Force required for equilibrium: $-3b\mathbf{j}$

5 f)

$\mathbf{P} + \mathbf{Q} = \begin{pmatrix} 2a \\ 6a \end{pmatrix} + \begin{pmatrix} b \\ 6b \end{pmatrix} = \begin{pmatrix} 2a+b \\ 6a+6b \end{pmatrix}$

Force required for equilibrium: $\begin{pmatrix} -2a-b \\ -6a-6b \end{pmatrix}$

6)

$a + 2b - 6b = 0$ and $3 - a + b = 0$

$a = 4$, $b = 1$

7 a)

$6\mathbf{i} + 10\mathbf{j} - 3.5\mathbf{i} - 12\mathbf{j} - 2.5\mathbf{i} = -2\mathbf{j}$

$a = 0$, $b = -2$

7 b)

2 N at a bearing of 180°

7 c)

The particle is accelerating due south.

7 d)

$2\mathbf{j}$ or 2 N due North

8)

$12\mathbf{i} + 11\mathbf{j} + 3\mathbf{i} - 15\mathbf{j} + a\mathbf{i} + b\mathbf{j}$

$= (15 + a)\mathbf{i} + (-4 + b)\mathbf{j}$

Due west: $b = 4$ and $a > -15$

9 a)

$\tan^{-1}\left(\frac{4}{3}\right) = 53.1°$ and $\tan^{-1}\left(\frac{12}{5}\right) = 67.4°$

Angle between: $14.3°$

9 b)

$\tan^{-1}\left(\frac{3}{2}\right) - \tan^{-1}\left(\frac{1}{5}\right) = 45°$

10)

$\binom{1}{-7} + \binom{k}{3k} = \binom{1+k}{-7+3k}$ parallel to $\binom{-4}{7}$

$\binom{1+k}{-7+3k} = p\binom{-4}{7}$

$1 + k = -4p$ and $-7 + 3k = 7p$

Solving simultaneously gives $k = \frac{21}{19}$

11 a)

Not moving east/west: $1 - 3 + a = 0 \Rightarrow a = 2$

Moving north: $2 + 8 + b > 0 \Rightarrow b > -10$

11 b)

Moving east: $1 - 3 + 8 = 6$

Moving North: $2 + 8 + b = 6 \Rightarrow b = -4$

12 a)

$\sqrt{3^2 + 4^2} = 5$ so $\frac{1}{5}\binom{3}{4}$

12 b)

$\sqrt{3^2 + 1^2} = \sqrt{10}$ so $\frac{1}{\sqrt{10}}(3\mathbf{i} - \mathbf{j})$

12 c)

$\sqrt{1^2 + 4^2} = \sqrt{17}$ so $\frac{1}{\sqrt{17}}\binom{-1}{4}$

12 d)

$\sqrt{8^2 + 6^2} = 10$ so $\frac{1}{10}(-8\mathbf{i} - 6\mathbf{j})$

13 a)

$\sqrt{3^2 + 3^2} = 3\sqrt{2}$ so $\mathbf{F} = \frac{1}{3\sqrt{2}}\binom{-3}{-3} \times 5 = \frac{5}{\sqrt{2}}\binom{-1}{-1}$

13 b)

$\binom{3}{0} - \binom{0}{-2} = \binom{3}{2}$

$\sqrt{3^2 + 2^2} = \sqrt{13}$ so $\mathbf{F} = \frac{1}{\sqrt{13}}\binom{3}{2} \times 4 = \frac{4}{\sqrt{13}}\binom{3}{2}$

13 c)

$\binom{5}{6} - \binom{1}{2} = \binom{4}{4}$

$\sqrt{4^2 + 4^2} = 4\sqrt{2}$ so $\mathbf{F} = \frac{1}{4\sqrt{2}}\binom{4}{4} \times 8 = \frac{1}{\sqrt{2}}\binom{8}{8}$

14 a)

$\sqrt{3^2 + 4^2} = \sqrt{25} = 5$ N

$F = ma \Rightarrow 5 = 0.5a \Rightarrow a = 10$ ms^{-2}

14 b)

$80 = 0 + \frac{1}{2} \times 10 \times t^2 \Rightarrow t = 4$ seconds

15 a)

$\sqrt{1^2 + 2^2} = \sqrt{5}$ N

$F = ma \Rightarrow \sqrt{5} = 2a \Rightarrow a = \frac{\sqrt{5}}{2} = 1.19$ ms^{-2}

15 b)

$v = 0 + \frac{\sqrt{5}}{2} \times 4 = 2\sqrt{5} = 4.47$ ms^{-1}

16 a)

$\binom{-2}{3} + \binom{5}{1} = \binom{3}{3}$ so $\sqrt{3^2 + 4^2} = 5$ N

$F = ma \Rightarrow 5 = 4a \Rightarrow a = 1.25$ ms^{-2}

16 b)

$10 = 0 + \frac{1}{2} \times 1.25 \times t^2 \Rightarrow t = 4$ seconds

17)

$\binom{5}{11.6} - 2\binom{0}{9.8} = \binom{5}{-8}$ so $\sqrt{5^2 + 8^2} = \sqrt{89}$ N

$F = ma \Rightarrow \sqrt{89} = 2a \Rightarrow a = 4.72$ ms^{-2}

18 a)

$\mathbf{F} = \binom{4.5}{13.6} - 2\binom{0}{9.8} = \binom{4.5}{-6}$

$\mathbf{F} = m\mathbf{a} \Rightarrow \binom{4.5}{-6} = 2\mathbf{a} \Rightarrow \mathbf{a} = \binom{2.25}{-3}$

$|\mathbf{a}| = \sqrt{2.25^2 + (-3)^2} = 3.75$ ms^{-2}

18 b)

$\mathbf{F} = \binom{4.5}{13.6} - 2\binom{0}{9.8} + \binom{-4.5}{5.5} = \binom{0}{-\frac{1}{2}}$

$\mathbf{F} = m\mathbf{a} \Rightarrow \binom{0}{-\frac{1}{2}} = 2\mathbf{a} \Rightarrow \mathbf{a} = \binom{0}{-\frac{1}{4}}$

$|\mathbf{a}| = \sqrt{0^2 + \left(-\frac{1}{4}\right)^2} = 0.25$ ms^{-2}

19)

$5 = 0 + \frac{1}{2}a \times 4 \Rightarrow a = 2.5 \text{ ms}^{-2}$

$\sqrt{4^2 + 3^2} = 5$ so $\frac{1}{5}\begin{pmatrix}4\\-3\end{pmatrix}$

$\mathbf{F} = m\mathbf{a} \Rightarrow \mathbf{F} = 4 \times 2.5 = 10 \text{ N}$

So $\mathbf{F} = \frac{1}{5}\begin{pmatrix}4\\-3\end{pmatrix} \times 10 = \begin{pmatrix}8\\-6\end{pmatrix}$

20)

$4 = 1 + a \times 5 \Rightarrow a = 0.6 \text{ ms}^{-2}$

$\sqrt{8^2 + 4^2} = 4\sqrt{5}$ so $\frac{1}{4\sqrt{5}}\begin{pmatrix}8\\4\end{pmatrix} = \frac{1}{\sqrt{5}}\begin{pmatrix}2\\1\end{pmatrix}$

$\mathbf{F} = m\mathbf{a} \Rightarrow \mathbf{F} = 3 \times 0.6 = 1.8 \text{ N}$

So $\mathbf{F} = \frac{1}{\sqrt{5}}\begin{pmatrix}2\\1\end{pmatrix} \times 1.8 = \frac{9\sqrt{5}}{25}\begin{pmatrix}2\\1\end{pmatrix}$

21 a)

$\begin{pmatrix}p\\q\end{pmatrix} + \begin{pmatrix}3\\-2\end{pmatrix} = \begin{pmatrix}p+3\\q-2\end{pmatrix}$ parallel to $\begin{pmatrix}2\\1\end{pmatrix}$

$1(p + 3) = 2(q - 2)$

$p + 3 = 2q - 4$

$p - 2q + 7 = 0$ as req.

21 b)

$p = 5 \Rightarrow q = 6$

$\begin{pmatrix}5+3\\6-2\end{pmatrix} = \begin{pmatrix}8\\4\end{pmatrix} \Rightarrow F = \sqrt{8^2 + 4^2} = 4\sqrt{5}$

$F = ma \Rightarrow 4\sqrt{5} = 5 \times a \Rightarrow a = \frac{4}{5}\sqrt{5} \text{ ms}^{-2}$

$s = 0 + \frac{1}{2} \times \frac{4}{5}\sqrt{5} \times 5^2 \Rightarrow s = 10\sqrt{5} = 22.4 \text{ m}$

22 a)

$(p + 3)\mathbf{i} + (q - 1)\mathbf{j}$ is parallel to $\mathbf{i} + 5\mathbf{j}$

$5(p + 3) = 1(q - 1)$

$5p + 15 = q - 1$

$5p - q + 16 = 0$ as req.

22 b)

$q = 6 \Rightarrow p = -2$

$\mathbf{i} + 5\mathbf{j} \Rightarrow F = \sqrt{1^2 + 5^2} = \sqrt{26}$

$F = ma \Rightarrow \sqrt{26} = 3 \times a \Rightarrow a = \frac{1}{3}\sqrt{26} \text{ ms}^{-2}$

$v = 0 + \frac{1}{3}\sqrt{26} \times 3 \Rightarrow v = \sqrt{26} = 5.10 \text{ ms}^{-1}$

23 a)

$\begin{pmatrix}p\\q\end{pmatrix} + \begin{pmatrix}-1\\4\end{pmatrix} = \begin{pmatrix}p-1\\q+4\end{pmatrix}$ parallel to $\begin{pmatrix}3\\4\end{pmatrix}$

$4(p - 1) = 3(q + 4)$

$4p - 4 = 3q + 12$

$4p - 3q - 16 = 0$ as req.

23 b)

$p = -2 \Rightarrow q = -8$

$\binom{-2-1}{-8+4} = \binom{-3}{-4} \Rightarrow F = \sqrt{3^2 + 4^2} = 5$

$F = ma \Rightarrow 5 = 2 \times a \Rightarrow a = \frac{5}{2}\ \text{ms}^{-2}$

$v = 2 + \frac{5}{2} \times 3 \Rightarrow v = 9.5\ \text{ms}^{-1}$